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And
MANAGEMENT DEVELOPMENT INSTITUTE OF SINGAPORE

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1. What is VPN?

A Virtual Private Network, constitutes an encrypted linkage via the internet, connecting a device to a network. This encrypted connection plays a vital role in ensuring the secure transmission of sensitive data, thereby thwarting unauthorized access and disruption in the traffic flow.

The genesis of VPN is intricately woven into the annals of technology history, particularly rooted in the United States. Notably, the pioneering VPN technology was conceptualized and executed by Microsoft. Indeed, Microsoft stood as an early innovator in VPN technology, primarily tailored for corporate applications. Historical accounts suggest that Gurdeep Singh-Pall might have been the mastermind behind the invention of VPN, a contention widely accepted.

The nascent VPN technologies of the mid-1990s were anchored in the Simple Peer to Peer Tunneling Protocol. As the mid-2000s unfolded, individual users began exhibiting heightened awareness regarding online and network security. Public networks, commonly found in cafes and airports, metamorphosed into breeding grounds for malicious cyber activities. By 2005, the inaugural commercial VPNs emerged in response to the escalating demand for user-friendly security solutions, streamlining the erstwhile intricate setup procedures.

The archetypal VPN protocol, the Point-to-Point Tunneling Protocol, remains deeply entrenched in contemporary usage. Recognized by its acronym PPTP, this protocol epitomizes the most rudimentary form of VPN service available. Functioning adeptly, PPTP establishes a secure and private network connecting two or more users across wide area network (WAN) or local area network (LAN) environments.

2. How does it work?

An IP address functions as a unique numerical identifier designated to each device, serving as a virtual point of reference for online interactions. The VPN acts as an intermediary for individuals accessing the internet, obscuring their IP address. This is precisely where the importance of the IP address becomes evident. Threat actors operating in the digital domain may potentially misuse an IP address using various methods. Disclosing an IP address can facilitate the commission of various illicit activities. Therefore, it is highly advisable to protect the confidentiality of one's IP address.

A VPN has the capability to conceal an IP address through the utilization of a VPN tunnel, thereby enhancing one's online privacy instantaneously. Upon connection to a VPN server, a new IP address is allocated to the device – one that is not owned by our internet service provider. Subsequent online actions post VPN server connection will be associated with the IP address of the VPN, rendering them untraceable back to the user.
The VPN tunnel establishes an encrypted link between our devices, such as smartphones, computers, and tablets, and the VPN server. It conceals our IP address and encrypts our data while traversing the internet, as well as the information generated during web browsing. Unauthorized entities are unable to access our online data or monitor our activities due to the impenetrable nature of the connection without a cryptographic key.

The types of tunnel protocols

1. WireGuard
2. NordLynx
3. OpenVPN
4. IKEv2/IPsec
5. L2TP/IPsec
6. SSTP
7. PPTP

2.1. Wire Guard

The protocol that demonstrates rapidity, especially beneficial in situations where speed is a priority, is highly commendable. Furthermore, it possesses a considerable level of security. The lightweight nature of WireGuard is noteworthy, attributed to its concise 4,000 lines of code, thereby reducing potential vulnerabilities and limitations. By virtue of being open source, it fosters transparency, enabling adaptability and simplifying the process of identifying and rectifying errors. WireGuard is still in its developmental stages and sets itself apart from OpenVPN and IPsec by mandating its distinct infrastructure for functionality.

2.2. Nord Lynx

In the year 2019, NordVPN introduced Nord Lynx, a newly developed protocol that combined the speed of WireGuard and enhanced it by bolstering user privacy and the high level of security desired by all.

2.3. OpenVPN

This protocol is characterized by its open-source essence and compatibility with a range of primary operating systems. The source code can be accessed for the purpose of examination and customization in alignment with individual preferences. OpenVPN

protocol exhibits the capacity to function over the TCP or UDP internet protocols, establishing itself as a highly secure VPN tunnel protocol renowned for its impressive speed. Notwithstanding the robust security attributes and effective operational efficiency, the configuration procedure for OpenVPN is widely acknowledged to be notably intricate.

2.4. IKEv2/IPsec

The security benefits offered by IPsec are accessible by employing the IKEv2/IPsec protocol, which integrates the swift performance linked with IKEv2. When faced with scenarios of VPN connection interruptions or network transitions, the automatic connection feature of IKEv2/IPsec effectively restores operations to their standard state. Unfortunately, some operating systems do not endorse the IKEv2/IPsec protocol, despite its praiseworthy characteristics.

2.5. L2TP/IPsec

The amalgamation of Layer 2 Tunneling Protocol and Internet Protocol Security, known as L2TP/IPsec, is utilized to create secure and encrypted connections over the internet. Widely utilized in VPN services, this protocol ensures confidentiality, integrity, and authentication of data transmission, providing strong security features for remote access to corporate networks.

2.6. SSTP

SSTP presents a straightforward setup process, accompanied by readily available support. This protocol ensures security and exhibits a relatively high speed, particularly effective in circumventing firewalls. The limitation, however, lies in its compatibility exclusively with Windows operating systems. Originating from Microsoft, this protocol is associated with the entity's established partnership with the NSA.

2.7. PPTP

PPTP, being expeditious and expedient for immediate VPN usage, exhibits high adaptability across various platforms and boasts user-friendly configuration and operation. This particular protocol is antiquated, consequently lacking in security measures and harboring various susceptibilities and weaknesses; it is subject to decryption by the NSA, albeit susceptible to firewall rejection.

The main process of VPN tunneling is to transmit data from end devices or network to another device or network and back without compromising the data privacy.

3. Hardware VPN

Hardware VPNs are specialized physical apparatus utilized for establishing secure connections between distant users and private networks. These solutions, encompassing routers, firewalls, and VPN concentrators, deliver robust encryption capabilities to safeguard data transmitted across the network. Particularly suited for

entities necessitating stringent security protocols, hardware VPNs guarantee the confidentiality and integrity of data, rendering them apt for managing delicate information and adhering to regulatory standards. Moreover, hardware VPNs offer scalability and adaptability, enabling organizations to effortlessly extend their VPN framework as required. Backed by their demonstrated dependability and comprehensive security functionalities, hardware VPNs persist as the favored option for enterprises in search of secure remote accessibility and site-to-site linkage.

4. Software VPN

Software VPNs provide virtualized solutions for establishing secure connections between remote users and private networks via the internet. In contrast to hardware VPNs, which depend on dedicated physical devices, software VPNs employ applications installed on user devices to oversee and encrypt network traffic. These solutions deliver flexibility and straightforward deployment, rendering them appropriate for organizations of all sizes. Furthermore, software VPNs can be seamlessly integrated with diverse operating systems and devices, facilitating smooth connectivity across multiple platforms. Despite potentially lacking the robust performance capabilities of hardware-based alternatives, software VPNs persist as a cost-efficient and scalable choice for organizations in search of secure remote access and network connectivity.

5. Cloud VPN

Cloud VPN serves as a fundamental element of contemporary networking solutions, bridging the connection between distant users and cloud-based assets through strong security measures and uninterrupted connectivity. Through the utilization of encryption protocols, Cloud VPN guarantees the confidentiality and protection of data transmitted via the internet, shielding it from potential risks. Its adaptability empowers entities to create secure links for remote entry, promoting effective cooperation among dispersed groups while upholding adherence to regulatory criteria. Additionally, Cloud VPN provides scalability and cost efficiency, permitting enterprises to adjust to changing network demands without incurring substantial infrastructure expenses. As an essential component of cloud computing frameworks, Cloud VPN seamlessly integrates with prominent cloud platforms such as AWS, Azure, and GCP, empowering entities to leverage the complete capabilities of cloud-native services while safeguarding confidential information.

6. Main Type of VPNs

1. Personal VPNs
2. Remote access VPNs
3. Mobile VPNs
4. Site to Site VPNs

6.1. Personal VPNs

In this service VPN gives encrypted access to each device individually to connect with third-party server. It will route your traffic to the server without showing anything about your identity and protect form surveillance. Also, can make access for you to bypass geo-blocking and circumvent firewalls in social media and online usage sensitively controlled countries.

6.2. Remote access VPNs

Remote Access VPN, establishes a secure conduit connecting a distant user's device to the internal network of the organization. Conceptualize it as a clandestine, encrypted passageway meticulously crafted within the vast and potentially perilous realm of the internet. Every piece of information conveyed through this passageway is enveloped in a cryptographic shield, indiscernible to prying individuals and resistant to covert surveillance, safeguarding your sensitive data from unauthorized intrusion.

6.3. Mobile VPNs

Mobile Virtual Private Networks (VPNs) broaden the features of standard VPNs for mobile devices, allowing secure access to private networks for users who are on the move. These solutions are crucial for off-site employees, allowing them to securely reach corporate assets from any location with internet access. Moreover, they provide functionalities like automatic connection management and smooth transition between Wi-Fi and cellular networks, ensuring uninterrupted connectivity for users. Mobile VPNs boost productivity and effectiveness for mobile workers while upholding adherence to security policies and regulations**.**

6.4. Site to Site VPNs

Site-to-site VPNs create secure connections between multiple remote networks, facilitating seamless communication over the internet. They are useful for different brunches of enterprise data bases connecting through the internet. By encrypting data transmission, site-to-site VPNs ensure confidentiality and security, enabling organizations to establish private network connectivity across distributed sites.

7. How will affect in security?

VPNs provide functionalities such as IP obfuscation and anonymization, concealing users' authentic IP addresses and geographic locations. This shield of anonymity poses challenges for websites, advertisers, or malevolent entities aiming to monitor users' online behaviors, thereby amplifying levels of privacy and security.

Besides, VPNs facilitate users in evading geographical constraints enforced by content providers or governmental entities, thereby accessing geo-blocked content and services securely. Additionally, VPNs streamline secure remote entry to corporate networks for employees stationed at home or on the move.

Despite the notable security advantages that VPNs offer, users are advised to opt for reputable VPN service providers, update their VPN software regularly, and adhere to optimal security protocols. Organizations, on the other hand, should institute comprehensive security measures to ensure the prudent and secure utilization of VPNs throughout their networks.

8. Network Legal Issues

Security breaches represent another noteworthy apprehension, considering that VPN servers and protocols could be susceptible to attacks, resulting in unauthorized access or data breaches. Strong security measures, routine software updates, and security evaluations are imperative to alleviate the danger of security breaches.

Compliance prerequisites differ based on jurisdiction and sector, encompassing laws on data protection, sector-specific regulations, and international agreements on data transfer. Failure to conform to these prerequisites may lead to legal repercussions, penalties, or harm to reputation.

Government surveillance and censorship endeavors also influence VPN utilization, with certain nations limiting or monitoring VPN traffic as part of their cybersecurity or national security endeavors. Users and organizations need to be cognizant of government regulations overseeing VPN utilization and evaluate the legal repercussions of employing VPNs in diverse jurisdictions.

Moreover, VPNs raise apprehensions regarding intellectual property rights, given their potential to bypass geographic constraints on copyrighted material. Content providers might deploy mechanisms to identify and obstruct VPN traffic to uphold licensing agreements and safeguard copyrighted content.

Jurisdictional concerns introduce an additional stratum of complexity, as VPN service providers function within varied legal frameworks featuring differing data protection standards. Grasping data sovereignty, legal jurisdiction, and avenues for redress in the event of disagreements or legal complications is vital for users and organizations leveraging VPN services.

In conclusion, effectively navigating network legal issues in VPNs necessitates a comprehensive comprehension of privacy legislation, security vulnerabilities, compliance requisites, and jurisdictional regulations. Organizations must enact stringent policies and protocols to tackle these issues and ensure the legal and ethical utilization of VPNs within their networks.

9. Remote System Administration

System administrators can manage servers, devices, and network infrastructure remotely from any location with internet connectivity. Through the creation of secure tunnels on public networks, VPN guarantees the secure execution of sensitive administrative activities, such as configuration, troubleshooting, and maintenance, thus safeguarding against unauthorized entry and data breaches. The significance of VPN in remote system administration transcends security, encompassing performance optimization as well. It enables administrators to reach resources situated in diverse geographical areas without physical presence, thereby streamlining administrative procedures and minimizing downtime. Moreover, VPN technologies provide functionalities such as split tunneling and Quality of Service (QoS) prioritization to enhance bandwidth utilization and ensure seamless performance for crucial administrative operations. In essence, VPN equips system administrators with the ability to effectively oversee and control remote systems, thereby boosting productivity, adaptability, and security within contemporary IT environments.

10. The Future of VPN

in the contemporary era of digitalization, the importance of VPN technology has significantly escalated, whether individuals aim to emphasize the protection of privacy and security, obtain entry to restricted content, or enable workforce mobility. Whether engaged in browsing from a local cafe or remotely retrieving sensitive work documents, a VPN guarantees the confidentiality of your data.

AI impacts every sector, including Virtual Private Networks (VPNs) in 2024, by improving routing, decreasing latency, and efficiently managing resources. It also forecasts server workloads, enhances traffic routing, performs behavioral analysis, and enhances encryption to boost VPN performance.

The significance of Quantum-safe VPNs lies in safeguarding data against quantum attacks, given that quantum computers have the capability to compromise conventional encryption techniques. Businesses allocate resources towards these VPNs to heighten security measures.

Diverging from conventional VPNs, Decentralized VPNs function via dispersed nodes, providing heightened security and user-driven nodes. This strategy diminishes the likelihood of failures and involves individuals contributing bandwidth to the network.

Edge computing VPNs handle data closer to its origin, emphasizing speed, latency reduction, and the establishment of secure connections in unreliable networks.

11. Conclusion

In conclusion, the evolution of VPN technologies has brought significant advancements in enhancing security, performance, and manageability in digital environments. Incorporating AI and quantum-safe mechanisms has further strengthened VPNs against emerging threats, ensuring robust protection for organizational networks and sensitive data.

However, alongside these technological advancements, legal and compliance considerations surrounding VPN use are paramount. A thorough understanding of privacy laws, security vulnerabilities, and jurisdictional regulations is essential to ethically and securely deploy VPN solutions. This knowledge enables organizations to navigate legal complexities and safeguard user privacy while leveraging VPNs effectively.

Various VPN solutions, including hardware, software, and cloud-based options, offer scalability, flexibility, and seamless integration with cloud services to meet diverse organizational needs. Innovations in VPN protocols like WireGuard and NordLynx prioritize speed, security, and user privacy, representing significant milestones in VPN technology.

Overall, the continuous evolution and innovation in VPN technologies provide organizations with powerful tools to enhance cybersecurity and operational efficiency. By addressing legal, compliance, and technological aspects comprehensively, organizations can harness the full potential of VPNs while maintaining ethical standards and ensuring secure digital environments.

12. Reflection

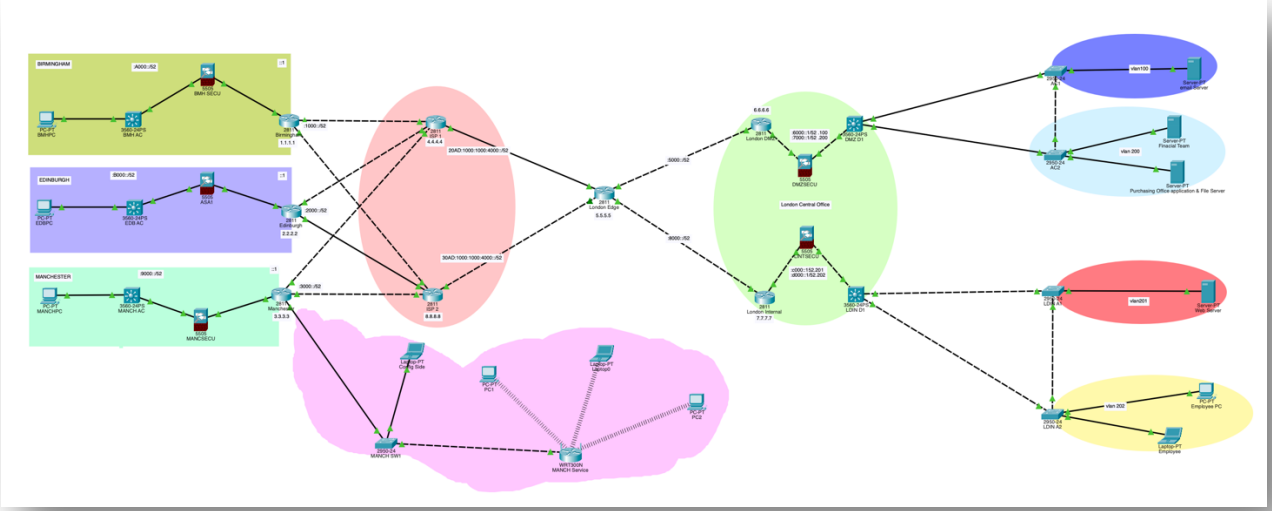
The development of VPN technology signifies a responsive approach to the constantly changing landscape of cybersecurity threats. The incorporation of AI and quantum-safe mechanisms demonstrates a proactive stance towards enhancing VPN security and performance, showcasing the industry's commitment to proactively address emerging risks and challenges.

Progress in VPN protocols such as WireGuard and NordLynx serves as an illustration of a forward-looking strategy in VPN advancement, emphasizing enhancements in speed, security, and user confidentiality. These developments exhibit a dedication to improving the overall VPN user experience and tackling key issues in contemporary networking environments.

Additionally, the cooperation between academic institutions and industry collaborators on VPN technology and management methodologies highlights the significance of scholarly contributions in propelling innovation and knowledge sharing. This partnership encourages ongoing enhancements and progression in the field, offering cutting-edge solutions and valuable insights into effective VPN implementation and management strategies for organizations.

On the whole, a contemplation of these progressions underscores the transformative influence of VPN technology on cybersecurity practices and the continuous endeavors to innovate and adjust to emerging challenges in digital security.

13. Evidence of labor for the Packet Tracer

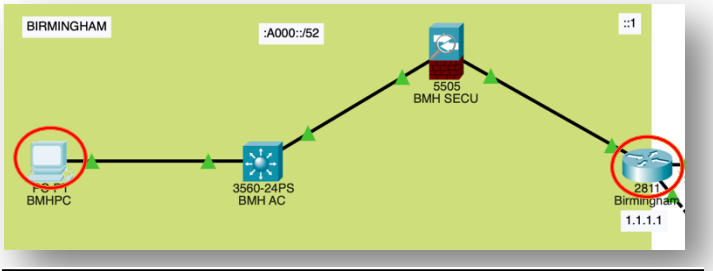


Identification of Router (Router – ID)

1. Birmingham Router – 1.1.1.1
2. Edinburg Router – 2.2.2.2
3. Manchester Router – 3.3.3.3
4. ISP 1 – 4.4.4.4
5. London Edge – 5.5.5.5
6. London DMZ – 6.6.6.6
7. London Internal 7.7.7.7
8. ISP 2 – 8.8.8.8

13.1. Ping Records

1. BMHPC - 20AD:1000:1000:A000::100/52 to BMH Router – 20AD:1000:1000:A000::1/52



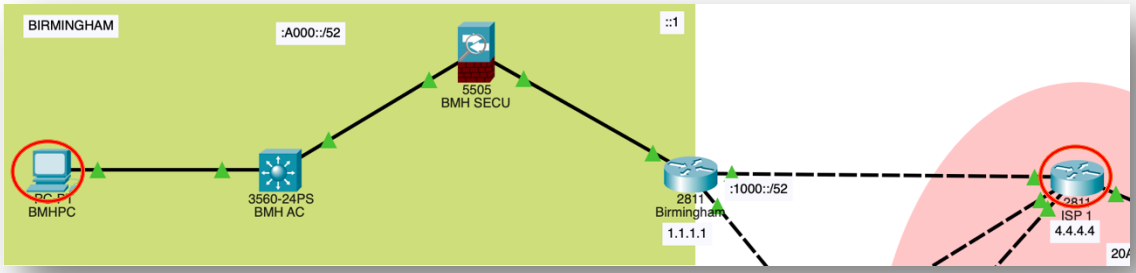

```
Cisco Packet Tracer PC Command Line 1.0
C:\>Ping 20AD:1000:1000:A000::1

Pinging 20AD:1000:1000:A000::1 with 32 bytes of data:

Reply from 20AD:1000:1000:A000::1: bytes=32 time<1ms TTL=255
Reply from 20AD:1000:1000:A000::1: bytes=32 time=13ms TTL=255
Reply from 20AD:1000:1000:A000::1: bytes=32 time<1ms TTL=255
Reply from 20AD:1000:1000:A000::1: bytes=32 time<1ms TTL=255

Ping statistics for 20AD:1000:1000:A000::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

2. BMHPC - 20AD: 1000:1000: A000::100/52 to ISP1 Router – 20AD: 1000:1000:1000::1/52



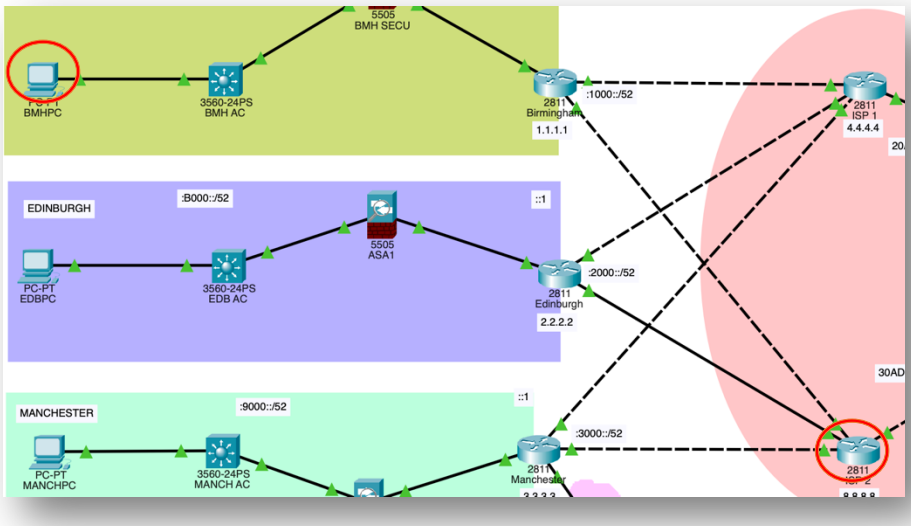

```
C:\>Ping 20AD:1000:1000:1000::1

Pinging 20AD:1000:1000:1000::1 with 32 bytes of data:

Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time=34ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254

Ping statistics for 20AD:1000:1000:1000::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 34ms, Average = 8ms
```

3. BMHPC - 20AD: 1000:1000: A000::100/52 to ISP2 Router – 30AD: 1000:1000:1000::1/52



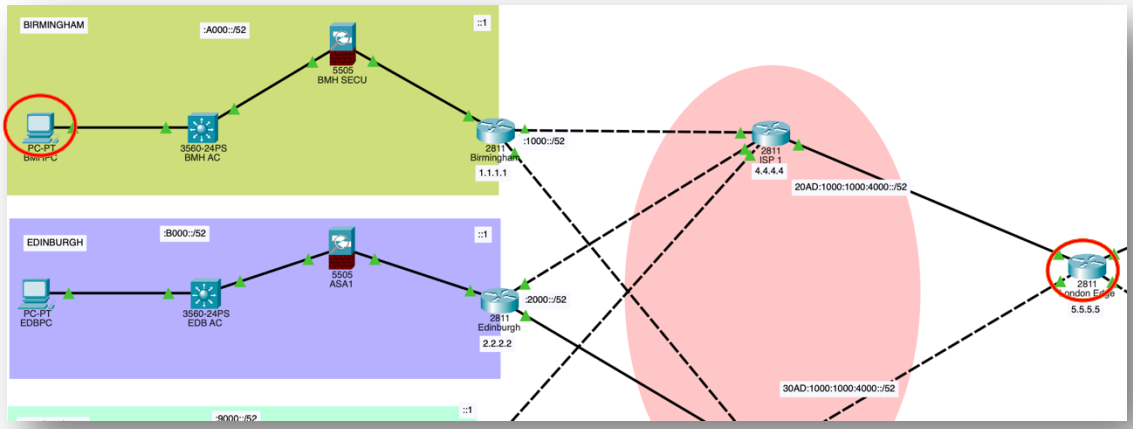

```
C:\>Ping 20AD:1000:1000:1000::1

Pinging 20AD:1000:1000:1000::1 with 32 bytes of data:

Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time=34ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254
Reply from 20AD:1000:1000:1000::1: bytes=32 time<1ms TTL=254

Ping statistics for 20AD:1000:1000:1000::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 34ms, Average = 8ms
```

4. BMHPC - 20AD:100 ...000:1000:4000::1/52



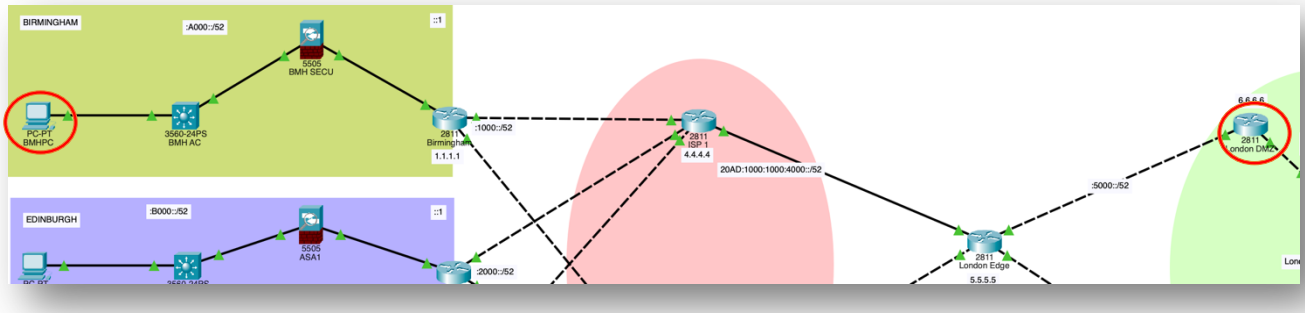

```
C:\>Ping 20AD:1000:1000:4000::2

Pinging 20AD:1000:1000:4000::2 with 32 bytes of data:

Reply from 20AD:1000:1000:4000::2: bytes=32 time<1ms TTL=253
Reply from 20AD:1000:1000:4000::2: bytes=32 time=1ms TTL=253
Reply from 20AD:1000:1000:4000::2: bytes=32 time=1ms TTL=253
Reply from 20AD:1000:1000:4000::2: bytes=32 time<1ms TTL=253

Ping statistics for 20AD:1000:1000:4000::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5. BMHPC - 20AD: 1000:1000:A000::100/52 to DMZ Router – 20AD:1000:1000:5000::1/52



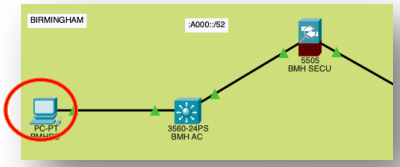

```
C:\>Ping 20AD:1000:1000:5000::2

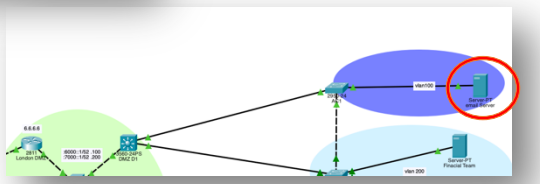
Pinging 20AD:1000:1000:5000::2 with 32 bytes of data:

Reply from 20AD:1000:1000:5000::2: bytes=32 time=1ms TTL=252
Reply from 20AD:1000:1000:5000::2: bytes=32 time=1ms TTL=252
Reply from 20AD:1000:1000:5000::2: bytes=32 time=74ms TTL=252
Reply from 20AD:1000:1000:5000::2: bytes=32 time=35ms TTL=252

Ping statistics for 20AD:1000:1000:5000::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 74ms, Average = 27ms
```

6. BMHPC - 2 000:6000::100/52





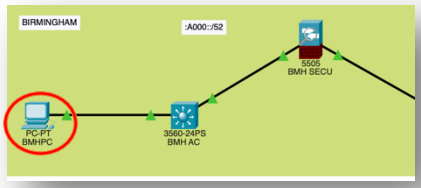

```
C:\>Ping 20AD:1000:1000:6000::100

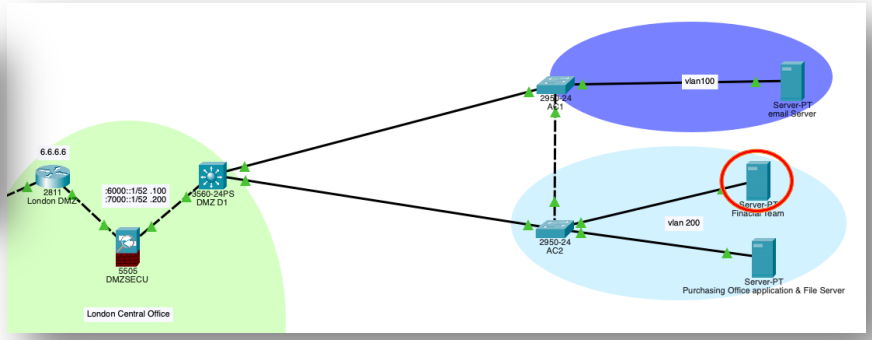
Pinging 20AD:1000:1000:6000::100 with 32 bytes of data:

Reply from 20AD:1000:1000:6000::100: bytes=32 time=2ms TTL=124
Reply from 20AD:1000:1000:6000::100: bytes=32 time=69ms TTL=124
Reply from 20AD:1000:1000:6000::100: bytes=32 time<1ms TTL=124
Reply from 20AD:1000:1000:6000::100: bytes=32 time=1ms TTL=124

Ping statistics for 20AD:1000:1000:6000::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 69ms, Average = 18ms
```

7. BMHPC - 20AD: 1000:1000: A000::100/52 to Financial - 20AD: 1000:1000:7000::200/52





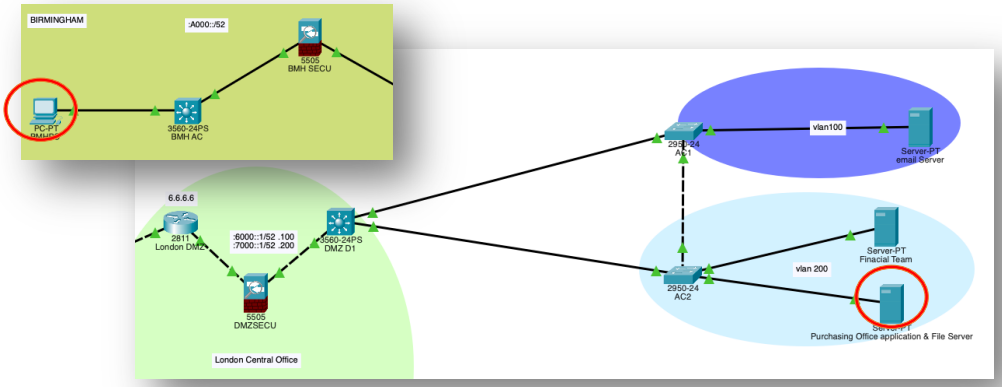

```
C:\>Ping 20AD:1000:1000:7000::200

Pinging 20AD:1000:1000:7000::200 with 32 bytes of data:

Reply from 20AD:1000:1000:7000::200: bytes=32 time=1ms TTL=124
Reply from 20AD:1000:1000:7000::200: bytes=32 time=2ms TTL=124
Reply from 20AD:1000:1000:7000::200: bytes=32 time<1ms TTL=124
Reply from 20AD:1000:1000:7000::200: bytes=32 time=48ms TTL=124

Ping statistics for 20AD:1000:1000:7000::200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 48ms, Average = 12ms
```

8. BMHPC - 2 000:6000::100/52



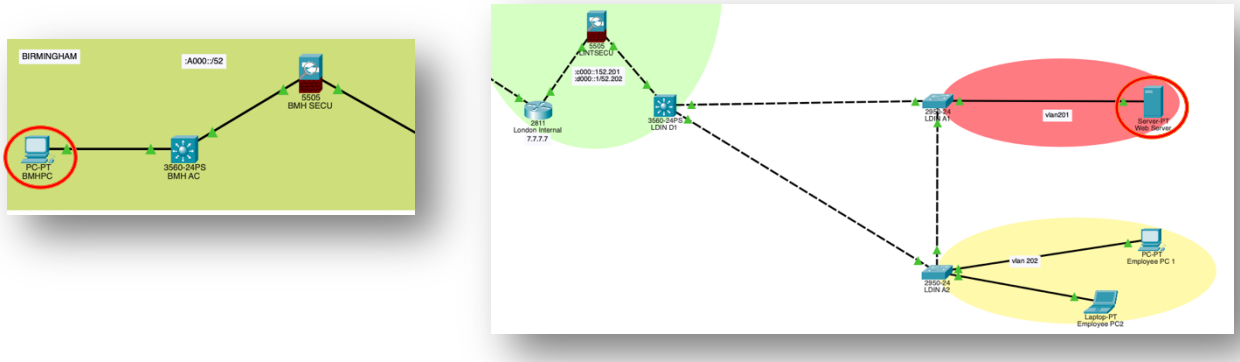

```
C:\>Ping 20AD:1000:1000:7000::203

Pinging 20AD:1000:1000:7000::203 with 32 bytes of data:

Reply from 20AD:1000:1000:7000::203: bytes=32 time=12ms TTL=124
Reply from 20AD:1000:1000:7000::203: bytes=32 time=46ms TTL=124
Reply from 20AD:1000:1000:7000::203: bytes=32 time=1ms TTL=124
Reply from 20AD:1000:1000:7000::203: bytes=32 time<1ms TTL=124

Ping statistics for 20AD:1000:1000:7000::203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 46ms, Average = 14ms
```

9. BMHPC - 20AD: 1000:1000: A000::100/52 to Web Server - 20AD: 1000:1000:C000::201/52



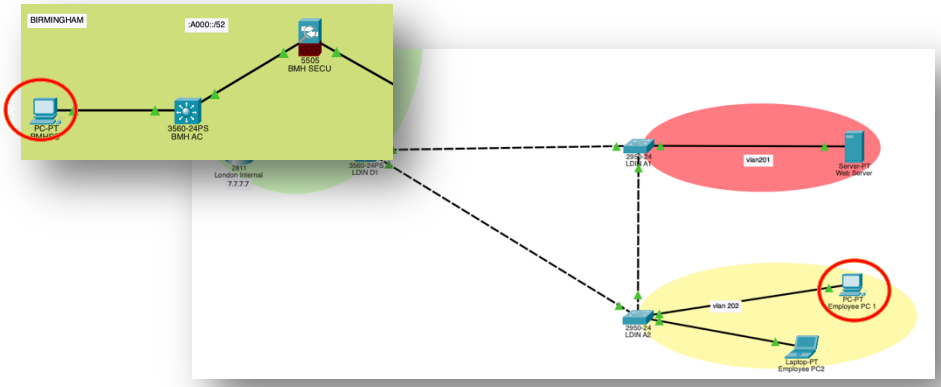

```
C:\>Ping 20AD:1000:1000:C000::201

Pinging 20AD:1000:1000:C000::201 with 32 bytes of data:

Reply from 20AD:1000:1000:C000::201: bytes=32 time=2ms TTL=124
Reply from 20AD:1000:1000:C000::201: bytes=32 time=44ms TTL=124
Reply from 20AD:1000:1000:C000::201: bytes=32 time=1ms TTL=124
Reply from 20AD:1000:1000:C000::201: bytes=32 time=13ms TTL=124

Ping statistics for 20AD:1000:1000:C000::201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 44ms, Average = 15ms
```

10. BMHPC - 2 : D000::202/52



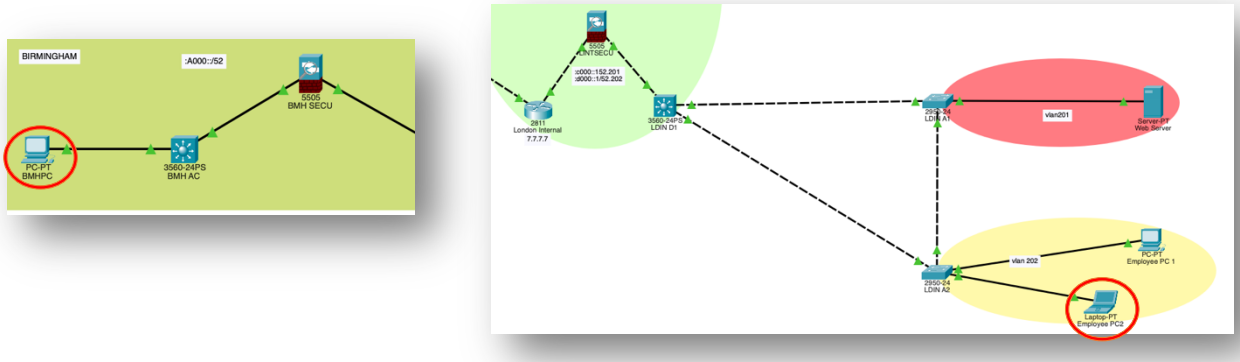

```
C:\>Ping 20AD:1000:1000:D000::202

Pinging 20AD:1000:1000:D000::202 with 32 bytes of data:

Reply from 20AD:1000:1000:D000::202: bytes=32 time<1ms TTL=124
Reply from 20AD:1000:1000:D000::202: bytes=32 time=63ms TTL=124
Reply from 20AD:1000:1000:D000::202: bytes=32 time=2ms TTL=124
Reply from 20AD:1000:1000:D000::202: bytes=32 time=54ms TTL=124

Ping statistics for 20AD:1000:1000:D000::202:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 63ms, Average = 29ms
```

11. BMHPC - 20AD: 1000:1000: A000::100/52 to EMP PC2 - 20AD: 1000:1000: D000::204/52



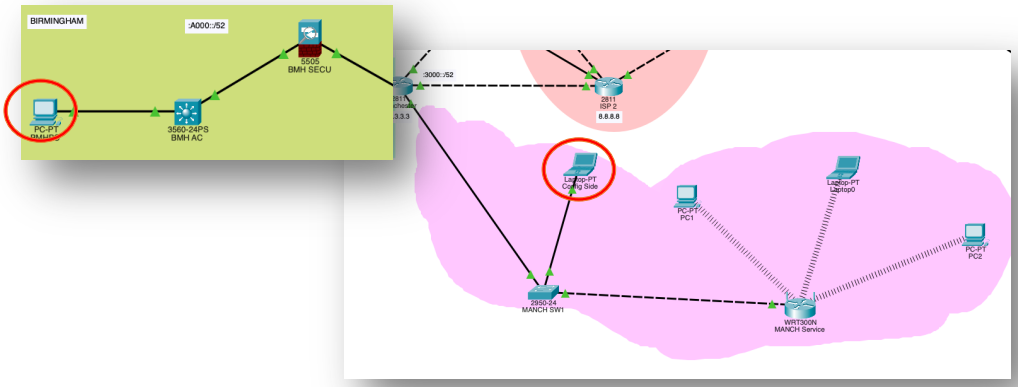

```
C:\>Ping 20AD:1000:1000:D000::204

Pinging 20AD:1000:1000:D000::204 with 32 bytes of data:

Reply from 20AD:1000:1000:D000::204: bytes=32 time=10ms TTL=124
Reply from 20AD:1000:1000:D000::204: bytes=32 time=79ms TTL=124
Reply from 20AD:1000:1000:D000::204: bytes=32 time<1ms TTL=124
Reply from 20AD:1000:1000:D000::204: bytes=32 time=1ms TTL=124

Ping statistics for 20AD:1000:1000:D000::204:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 79ms, Average = 22ms
```

12. BMHPC - 2 0: D000::202/52

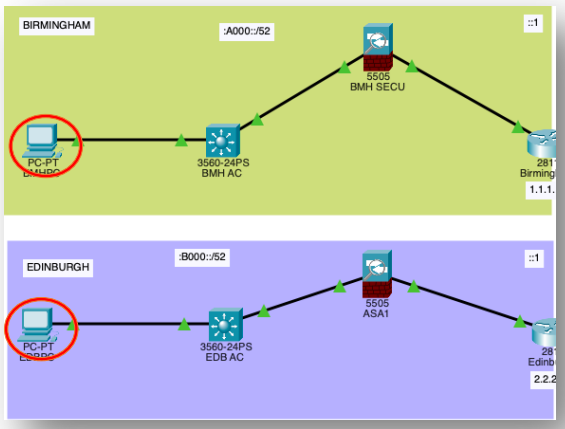



```
Pinging 20AD:1000:1000:F000::100 with 32 bytes of data:

Reply from 20AD:1000:1000:F000::100: bytes=32 time=1ms TTL=125
Reply from 20AD:1000:1000:F000::100: bytes=32 time=1ms TTL=125
Reply from 20AD:1000:1000:F000::100: bytes=32 time=31ms TTL=125
Reply from 20AD:1000:1000:F000::100: bytes=32 time<1ms TTL=125

Ping statistics for 20AD:1000:1000:F000::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 31ms, Average = 8ms
```

13. BMHPC - 20AD: 1000:1000: A000::100/52 to EDBPC - 20AD: 1000:1000: B000::204/52



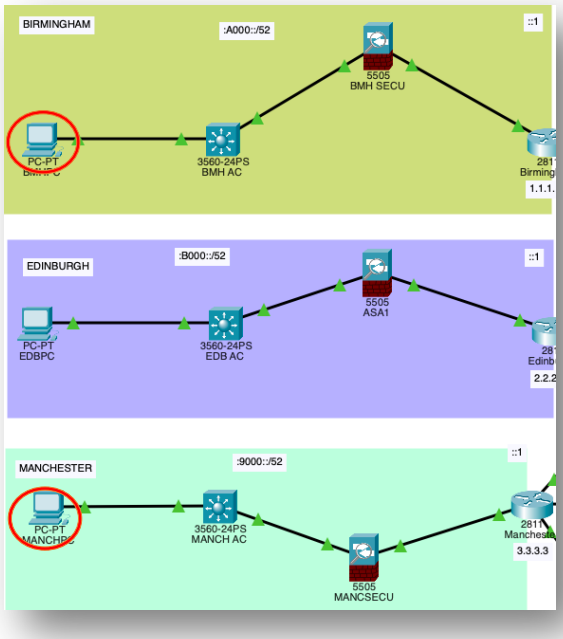

```
C:\>Ping 20AD:1000:1000:B000::100

Pinging 20AD:1000:1000:B000::100 with 32 bytes of data:

Reply from 20AD:1000:1000:B000::100: bytes=32 time=1ms TTL=125
Reply from 20AD:1000:1000:B000::100: bytes=32 time=78ms TTL=125
Reply from 20AD:1000:1000:B000::100: bytes=32 time=1ms TTL=125
Reply from 20AD:1000:1000:B000::100: bytes=32 time=9ms TTL=125

Ping statistics for 20AD:1000:1000:B000::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 78ms, Average = 22ms
```

14. BMHPC - 20A 00::100/52




```
Cisco Packet Tracer PC Command Line 1.0
C:\>Ping 20AD:1000:1000:9000::100

Pinging 20AD:1000:1000:9000::100 with 32 bytes of data:

Reply from 20AD:1000:1000:9000::100: bytes=32 time=24ms TTL=125
Reply from 20AD:1000:1000:9000::100: bytes=32 time=44ms TTL=125
Reply from 20AD:1000:1000:9000::100: bytes=32 time=65ms TTL=125
Reply from 20AD:1000:1000:9000::100: bytes=32 time=83ms TTL=125

Ping statistics for 20AD:1000:1000:9000::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 83ms, Average = 54ms
```

13.2. Routers & Switches Configuration (Packet Tracer)

#show run

Birmingham Router ID – 1.1.1.1

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:A000::1/52
 ipv6 address 30AD:1000:1000:A000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 description to ISP 1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:1000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:1000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
```

Edinburg Router ID – 2.2.2.2

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:B000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:2000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 description to ISP 1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:2000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
```

Manchester Router ID – 3.3.3.3

```
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:3000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1.88
 no ip address
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
!
router rip
!
ipv6 router ospf 1
 router-id 3.3.3.3
 log-adjacency-changes
```

ISP1 Router ID – 4.4.4.4

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:4000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:1000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:2000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:3000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
'
```

London Edge ID – 5.5.5.5

```
interface FastEthernet0/0
 description to ISP 1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:4000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:5000::1/52
 ipv6 address 30AD:1000:1000:5000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:8000::1/52
 ipv6 address 30AD:1000:1000:8000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:4000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
```

London DMZ ID – 6.6.6.6

```
 no ip address
 ipv6 address 20AD:1000:1000:6000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/0.200
 encapsulation dot1Q 200
 no ip address
 ipv6 address 20AD:1000:1000:7000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:5000::2/52
 ipv6 address 30AD:1000:1000:5000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 6.6.6.6
 log-adjacency-changes
```

London Internal ID – 7.7.7.7

```
.
interface FastEthernet0/0.201
 encapsulation dot1Q 201
 no ip address
 ipv6 address 20AD:1000:1000:C000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/0.202
 encapsulation dot1Q 202
 no ip address
 ipv6 address 20AD:1000:1000:D000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 20AD:1000:1000:8000::2/52
 ipv6 address 30AD:1000:1000:8000::2/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 7.7.7.7
 log-adjacency-changes
```

ISP 2 Router ID – 8.8.8.8

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:4000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:2000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/0
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:1000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface FastEthernet1/1
 no ip address
 duplex auto
 speed auto
 ipv6 address 30AD:1000:1000:3000::1/52
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan1
 no ip address
 shutdown
!
ipv6 router ospf 1
 router-id 8.8.8.8
 log-adjacency-changes
```

London DMZ Multilayer switch

```
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

London DMZ Switch AC1

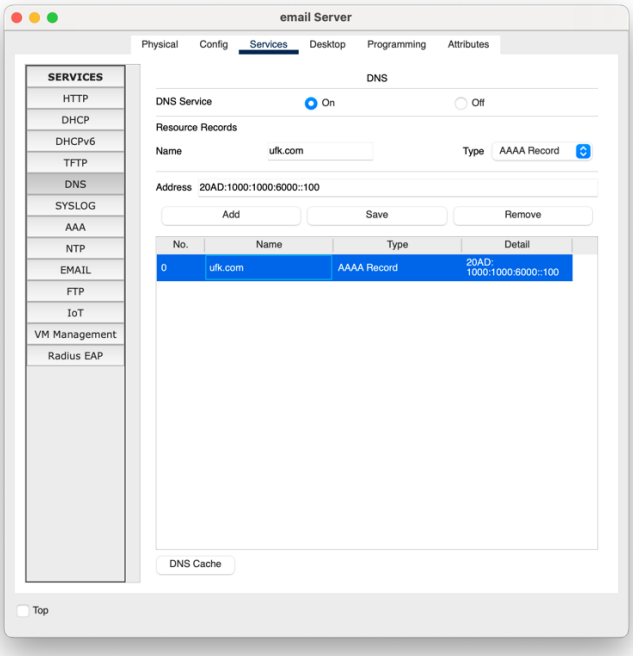
```
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 100
 switchport mode access
```

London DMZ Switch AC2

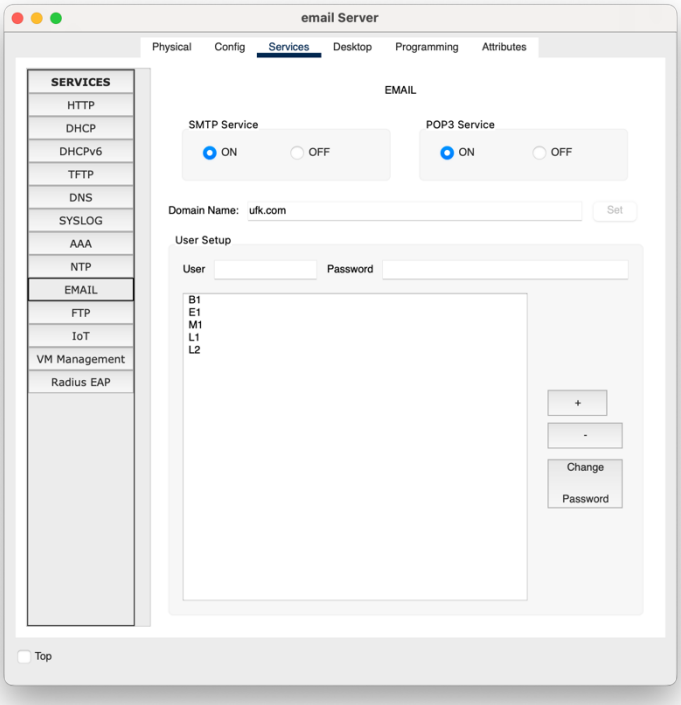
```
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface FastEthernet0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

13.3. Email Server Configuration (Packet Tracer)

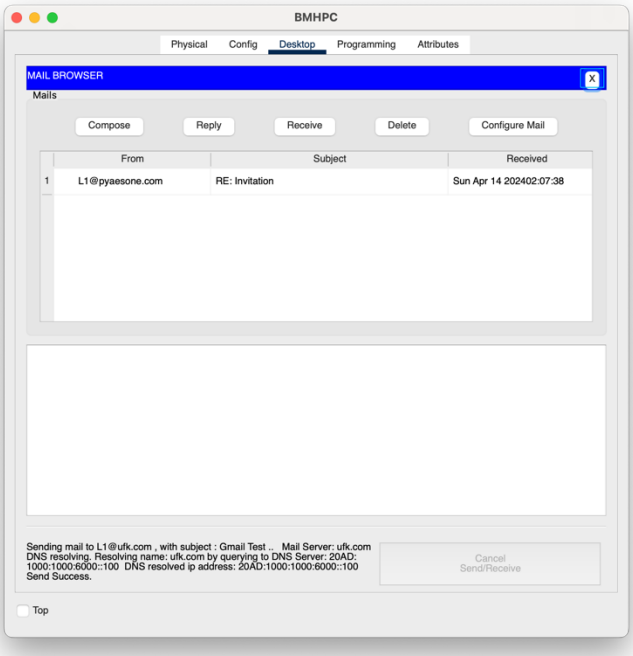
- DNS Service Config
- Domain name – ufk.com
- AAAA Record Type for Ipv6

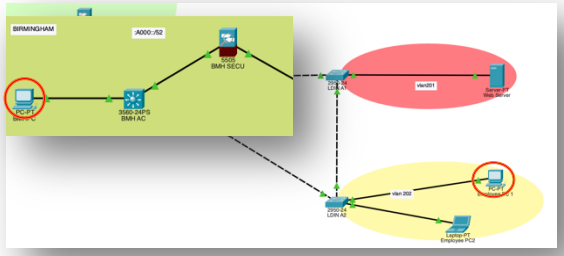


- Birmingham_ B1@ufk.com
- Edinburg_ E1@ufk.com
- Manchester_ M1@ufk.com

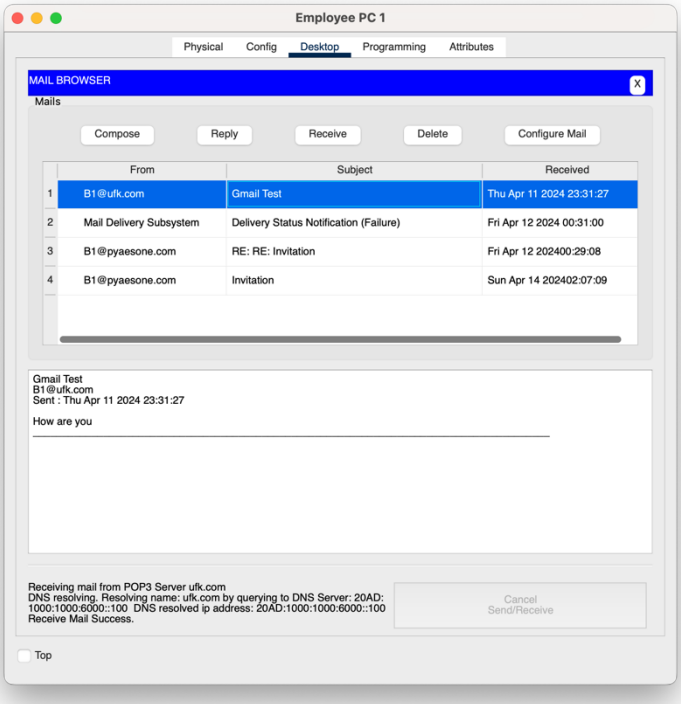


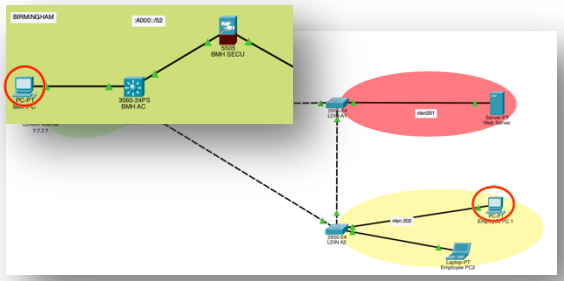
- BMHPC Sent an email to London Internal Employee PC1
- Username _ B1@ufk.com to L1@ufk.com
- Domain name – ufk.com





- London Internal Employee PC1 received an email from BMHPC
- Domain name – ufk.com





14. References

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https://www.archonsecure.com/blog/hardware-vpn-vs.-software-vpn

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